



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

SECP2523 - DATABASE

SEM 1 - 2025/ 2026

INITIAL SYSTEM PROPOSAL

HASTA TRAVEL VEHICLE RENTAL

MANAGEMENT SYSTEM

Group Name: DaTeam

Members:

- 1. MUHAMMAD ADAM ASHRAFF BIN ZAMRI (GROUP LEADER)**
- 2. MIKAEL HAQIMI BIN NAHAR JUNAIDI**
- 3. AHMAD WILDAN BIN MAZANI**
- 4. JOSHUA LING SHANG YANG**

Stakeholder: HASTA TRAVEL & TOURS SDN. BHD

Representative Name	Date Interviewed
1. ALIF FIRDAUS BIN JALALUDIN	28 th OCTOBER 2025
2. MUHAMMAD FAIZ IRFAN BIN ARIFFIN	28 th OCTOBER 2025
3. FARHAN SHAZWAN SHAH BIN SHAHBANI	28 th OCTOBER 2025

PROPOSED SYSTEM BACKGROUND

1. Introduction

Our academic project involves digital transformation in a Work-Based Learning (WBL) partnership with Hasta Travel & Tours Sdn. Bhd. (Hasta Travel), a car rental company operating at Universiti Teknologi Malaysia (UTM). Although the company has an online booking system in place, most core business processes, including customer data management, handling deposits, and tracking the status of vehicles, are still done manually. This project aims to overcome these issues by analyzing the existing processes and developing a central system that meets contemporary automated business standards.

The current manual process has resulted in high operational friction, including booking conflicts, delayed responses from customers, and challenges in confirming payment receipts. Without using an integrated system, the team has to deal with repetitive tasks in different systems, which increases the possibility of human error and makes it difficult to coordinate the process. These issues show the need for a more organized system to ensure accuracy and a seamless experience for both the team and the customer.

To counter these problems, we suggest a device-independent, integrated management solution consisting of four primary modules: User, Booking, Fleet, and Reporting. This solution will enable customers to search for cars and process payments effortlessly, while also offering the staff a separate interface to track the status of cars in real-time and confirm transactions. By integrating all major activities under one roof, this solution is expected to minimize manual intervention and help Hasta Travel adapt to a more competitive, automated environment.

2. Background Study

2.1 Organizational Background

Hasta Travel & Tours Sdn. Bhd. is a licensed service provider for travel and transportation services based in Skudai, Johor. The organization founded by Nabihah Kamarudin alumni of UTM and began as a student movement in August 2018. However, the organization has grown and become a significant contributor to the Universiti Teknologi Malaysia community. Hasta Travel is currently located at the Universiti Teknologi Malaysia Student Mall and provides services in the following areas: car rentals, travel agency services, and logistics.

Hasta Travel is designed to accommodate students, lecturers, and the general public. They also offer car rental services with their array of vehicles, ranging from economical hatchback cars such as Perodua Axia/Myvi to sedans, SUVs, and MPVs.

The mission of Hasta Travel is to facilitate convenient transportation services for UTM students and travelers in Johor by offering cheap and safe transportation. And the vision is to be one of the leading student-centric travel service brands in Malaysia by filling the gap between academic and professional travel services. Their promise is to deliver safety, excellence in vehicle maintenance, and support for local university programs.

2.2 Benefits of the Proposed System

2.2.1 Efficient in Operation and Automation

- **Centralized Work Environment**

The team can save time spent using split workflow by centralizing operation into one platform.

- **Human Error Reduction**

Automating and recording any data which minimizes manual chat with customer customers.

- **Real-Time Fleet Inventory**

The dashboard in this system updates real time information about vehicle availability to avoid conflicts like double booking issues.

2.2.2 Customer Experience Improvement

- **Smooth Booking Process**

Customers do need to wait for Whatsapp replies or any manual process because the system helps customers to search, filter and book any vehicles at any time.

- **Easy Payments**

The QR code for payment is implemented and receipt is generated to provide a modern and smooth payment experience.

- **Direct Communication**

Agreement in digital form and automated status updates (verification and booking confirmation) make sure the customer is always informed.

2.2.3 Business Growth & Scalability

- **Large Data Handling**

The system can scale up to 30,000 as potential users at UTM approximately.

- **Loyalty & Retention**

Loyalty and Rewards feature will attract users to use this system.

- **Decision on Data-Driven Analysis**

With the analysed reports gained, the admin or business owner can analyze revenue trends and demand periods to optimize pricing for business expansion.

3. Problem Statement

3.1 Heavy Reliance on WhatsApp

Hasta Travel still depends a lot on WhatsApp for booking communication. Even though customers fill in a form that sends their details to WhatsApp automatically, staff must handle the rest of the process manually through chat. This causes slow confirmations, inconsistent communication, and delays when staff are unavailable.

3.2 Limited Real-Time Management in DriveO

DriveO is an online system that provides minimal control measures for the staff. Bookings cannot be processed in real time since the staff is still expected to confirm the vehicle availability. Usually, this causes more confusion and even booking of a single vehicle by multiple customers simultaneously.

3.3 Lack of Important Features

The company has customer profiles, but there is basic customer information, which does not include any loyalty or reward programs. Only notifications are sent via WhatsApp, which may be missed by the customer. Data security is low, and the reports provided by the system are not complex enough to analyze the performance of the system.

3.4 The workflow is split between two platforms

Staff must switch between WhatsApp (for customer communication) and Driveo (for record keeping). This split workflow slows down operations, makes coordination harder and increases the chance of human error.

3.5 Poor Accuracy and Customer Experience

The current system makes it hard to maintain accurate information, respond quickly to customers, and track vehicles efficiently. As a result, customers do not receive a smooth and professional experience during the booking process.

3.6 The process does not scale well for a large target market.

With an estimated 30,000 potential users (mainly university students), the current WhatsApp-first workflow will become overwhelming, making it hard to respond quickly and maintain high service quality.

4. Proposed Solutions

4.1 Proposed Solution

The proposed solution for Hasta Travel is a centralized and device-agnostic vehicle rental management system that aims to automate existing manual processes and provide a seamless digital experience for users. By integrating various business operations into one system, the proposed system aims to address existing business operations and provide better services to the UTM community.

4.1.1 Digital Lifecycle Management

The system will manage the entire lifecycle of the rental process, moving away from fragmented manual entries towards a smooth and synchronized digital lifecycle. This entire change can be divided into three main phases:

- **Pre-Rental & Verification**

The system will introduce a secure and digital onboarding process, where customers will have the ability to upload their identity documents (Student ID and Driving License). The system will then provide automatic updates when the staff verifies the customer's profile.

- **Booking & Financial Automation**

Customers will have the ability to search and filter the vehicle according to the availability. The system will provide cost estimates instantly and will also manage the deposit pre-authorizations. The system will also provide a digital agreement and will allow the customer to pay through QR code transfer or online transfer.

- **Post-Rental & Loyalty**

Once the customer returns the vehicle, the system will calculate the final charges and will automatically deduct the late return fees, if applicable. To ensure customer retention, the system will provide a loyalty program that will allow customers to earn points on every successful rental.

4.1.2 Enhanced Operational Control

For the staff and the administrators, the solution offers control at the highest level through the provision of a dedicated management dashboard.

- **Fleet & Maintenance Tracking**

The solution tracks the miles covered by the vehicles, eliminating the need for physical logs, with the ability to send reminders for maintenance.

- **Role-Based Management**

The solution offers access control, thereby enhancing data security. The staff are able to concentrate on the day-to-day activities, such as verification of documents and inspection of vehicles, while the Admin can access the financial reports, demand trends, and risk management tools, including the ability to blacklist clients.

- **Quality Assurance**

The provision of the feedback and rating tool enables the company to track the quality of service provided, thereby enhancing data-driven improvements based on actual customer feedback.

4.2 Feasibility Study

4.2.1 Technical Feasibility

The proposed system is technically feasible given the available web development tools and the use of RDBMS technology. The developer group within the committee is deemed to have enough exposure to the basics of database concepts, system analysis, and even the basics of web development tools. In terms of hardware feasibility, available hardware, such as personal computers and internet connectivity, will suffice for the development of the system. Therefore, there is no need to acquire special hardware for the development of the system.

4.2.2 Operational Feasibility

From an operational perspective, the proposed system will therefore be feasible as it will match the existing operation flow of Hasta Travel, given that the office is adopting digital platforms such as DRIVEO and WhatsApp, which signify that the office will have a central system in place. The proposed system will ease day-to-day activities, as it will reduce errors, ease coordination between the office, customers, and crew, as minimal training will be needed, considering its user-friendly nature.

4.2.3 Economic Feasibility

The system is economically feasible as it is part of an academic project and as such requires minimal cost of developing it. The organization will also benefit as it will reduce inefficiencies in business operations and errors in bookings and data management without incurring higher costs of obtaining software to run it. In that respect, it also has long-term benefits of reducing labor and increasing quality of service.

5. Objectives

The development of the proposed vehicle rental management system is driven by the primary mandate of modernizing and automating the existing manual workflow. A key requirement of the system is the need to streamline the booking process by incorporating an automated processing engine, which allows the booking process to be speeded up and automated within the system. In line with the automation of the workflow, the system also incorporates the need to automate the generation of financial documentation, such as the generation of payment receipts and the automation of the process of sending the receipt via social media platforms such as email or WhatsApp. Additionally, the system incorporates the need to ensure high integrity of the data by ensuring that all the information and data are automatically saved to the database within the system.

In addition to the technical automation, the system aims to significantly enhance the service system for the Universiti Teknologi Malaysia (UTM) community. Among the major priorities is to ensure that the services are developed to be better, safer, and more convenient for the students and staff of UTM, offering a more satisfying and comfortable vehicle rental service to the customers. This is to ensure that the customer is able to have a safe, convenient, and impressive experience from the beginning to the end of the vehicle rental process. Ultimately, through the centralization of these services, it is possible for the staff and admins to manage the system more efficiently and in a more organized manner, providing a professional working environment with excellent user satisfaction.

6. Scopes

6.1 System Scope

We received clarification from Hasta Travel that the new system will focus on the vehicle rental service rather than its travel business or other businesses. This will help us to focus on the rental service project optimally. We will cover crucial parts which are vehicle booking, tracking availability, handling payments and managing customers.

Additionally, this system will specifically focus on daily challenges in vehicle rental operations like real-time status updates, confirming bookings automatically and quick response between customers and staff.

6.2 System Boundary

The system's boundaries define what **is included** and **not included** within the current development scope. For this project, the system will **only cover the vehicle rental operations** of Hasta Travel.

Included within the system boundary:

- Customer registration
- Vehicle listing and availability management
- Booking and rental confirmation process
- Payment recording or tracking
- Vehicle return and status updates
- Basic customer profile management

Excluded (outside the system boundary):

- GPS tracking and live vehicle monitoring
- Integration with third-party travel platforms (DRIVEO)

By setting up these boundaries, the project team should remain focused on the development of a complete and reliable car rental management system, without overcomplicating the current phase. This will further ensure that future expansion can be carried out more effectively when the car rental system is fully established.

7. Project Planning

7.1 Human Resource

Table 1: Role Allocation and Responsibility

Name	Role	Responsibility
MUHAMMAD ADAM ASHRAFF BIN ZAMRI	Project Leader	Plan and coordinate project activities, keep an eye on progress, make sure tasks are done on time, and make sure the project's goals are met.
MIKAEL HAQIMI BIN NAHAR JUNAIDI	System Analyst	Analyse current system, gather requirements, and document business processes
AHMAD WILDAN BIN MAZANI	Database Designer / Developer	Design database structure and develop system modules
JOSHUA LING SHANG YANG	Documentation & Testing	Prepare project documentation and conduct system testing

7.2 Work Breakdown Structure (WBS)

Level 1: Hasta Vehicle Rental Management System Development

Level 2: Project Components

1. Hasta Vehicle Rental Management System (Main Product)

1.1 User Access & Authentication

- 1.1.1** Registration & Login (Email/Phone-based authentication)
- 1.1.2** Identity Verification System (Student ID & License upload/approval)
- 1.1.3** Role-Based Access Control (Customer, Staff, and Admin panels)
- 1.1.4** Profile Management (Password reset and personal data updates)

1.2 Booking Management Module

- 1.2.1** Real-Time Vehicle Search & Filtering (Type, Date, Location)
- 1.2.2** Online Booking Engine (Availability check and pending record creation)
- 1.2.3** Booking Status & History Tracking (Active, Completed, Cancelled)
- 1.2.4** Cancellation & Modification Feature (Date/Vehicle updates and refund triggers)

1.3 Payment & Deposit Management

- 1.3.1** Payment Gateway & QR Integration (Online transfer and receipt upload)
- 1.3.2** Automated Deposit Handling (Pre-authorization and release logic)
- 1.3.3** Billing & Penalty System (Late return fees and custom charges)
- 1.3.4** Digital Receipt & Agreement Generation

1.4 Vehicle & Inventory Control

- 1.4.1** Real-Time Vehicle Availability Dashboard (Available, Rented, Maintenance)
- 1.4.2** Vehicle Category Management (Make, Model, Rate configuration)
- 1.4.3** Maintenance Scheduling (Mileage-based alerts and usage logs)

1.5 Operational & Monitoring Features

1.5.1 Vehicle Handover & Return Workflow (Condition documentation/photo upload)

1.5.2 Rental Timer & Overdue Detection

1.5.3 Reward & Loyalty Points System (Tier assignment and rules)

1.5.4 Blacklist & Risk Management (Flagging problematic/fraudulent users)

1.6 Reports & Admin Analytics

1.6.1 Revenue & Financial KPI Reporting

1.6.2 Booking Trend Analysis (Weekly/Monthly demand visualization)

1.6.3 Operational Reports (Daily Pickup/Return schedules)

1.7 Quality Assurance & Data Security

1.7.1 Secure Database Management (Encrypted user data)

1.7.2 Full-System Integration Testing (All modules end-to-end)

1.7.3 User Acceptance Testing (UAT) and Deployment

There are seven components in the product-oriented WBS of the Hasta Travel Vehicle Rental Management System as described above. Each component represents an important part of the system, ranging from user authentication to automated booking, until monitoring and analytics. This detailed breakdown of the system means that every aspect, from initial document verification to the final awarding of loyalty points, is covered with clarity. This breakdown of the system into work packages means that it is easy to distribute tasks among the members of the development team, ensuring that progress is monitored with accuracy throughout the development process.

7.3 Gantt Chart

Github link: [Gantt Chart](#)

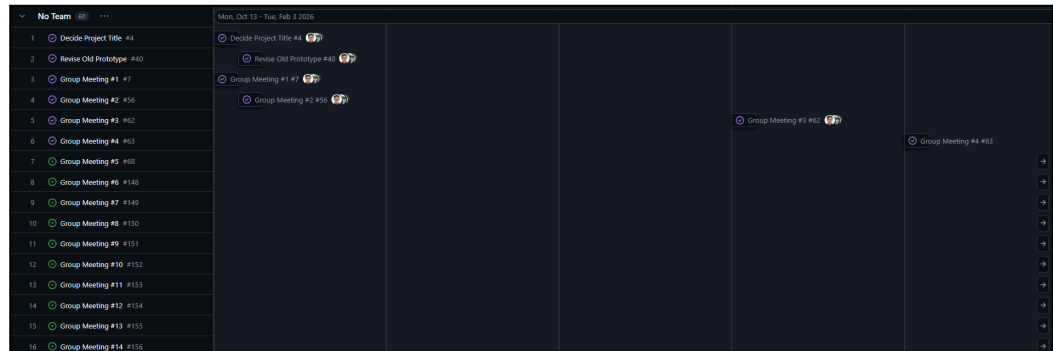


Figure 1: Gantt Chart Part 1

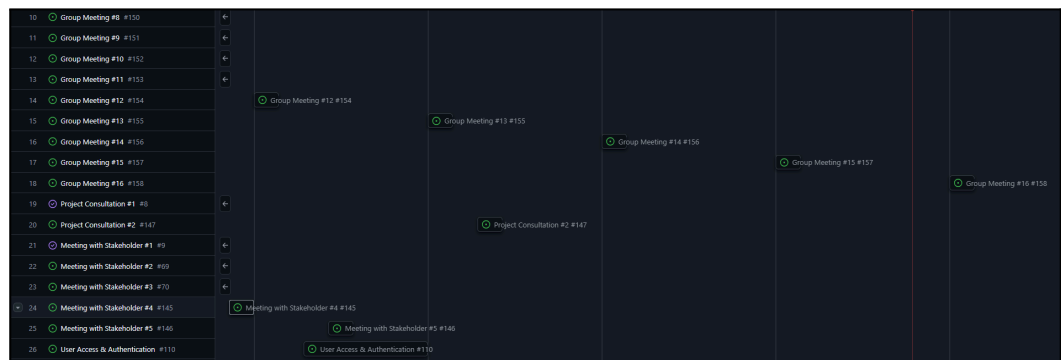


Figure 2: Gantt Chart Part 2

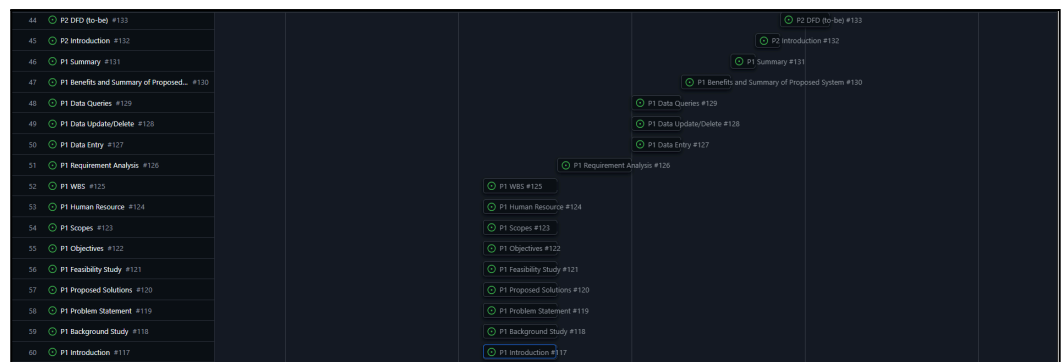


Figure 3: Gantt Chart Part 3

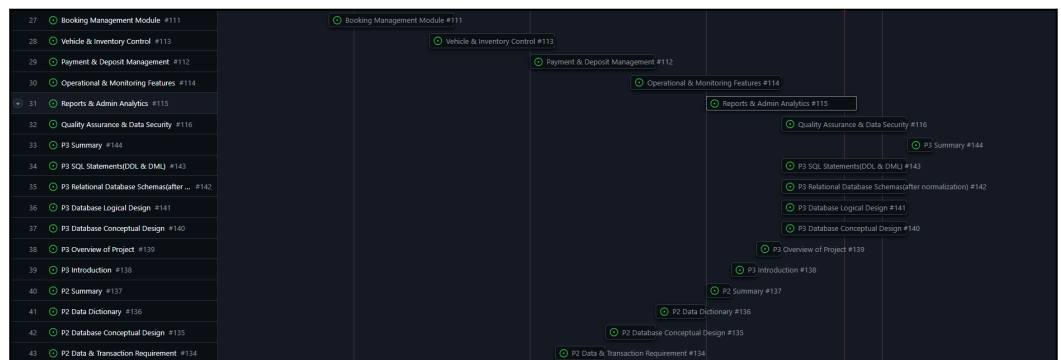


Figure 4: Gantt Chart Part 4

8. Requirement Analysis (AS-IS System)

8.1 Current Business Process

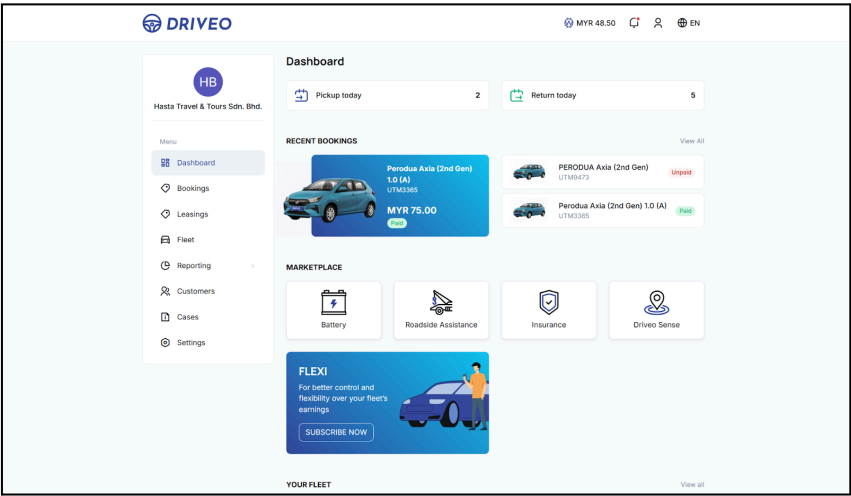


Figure 5: Current System’s Dashboard

Figure 5 above shows the dashboard of the staff & admin management page from DRIVEO system. In the current system, staff could check the pickup and return car status and information for that present date. Besides, they can also view the bookings recently with the payment status according to each booking.

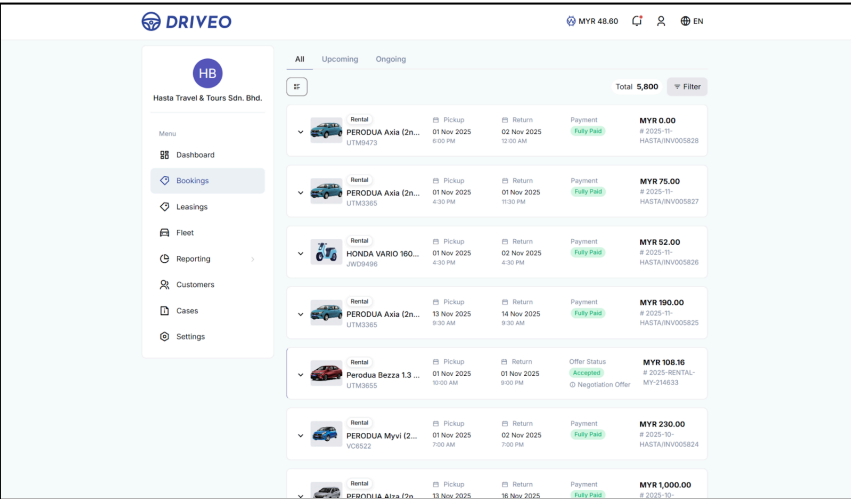


Figure 6: Booking Page of Current System

In **Figure 6**, it is the vehicle rental bookings page to check for all the requirements. Each booking will show the car information, pickup date, return date, payment status and invoice with the payment amount. To search the order looking for, staff can filter the statement according to payment status, sorting, status, sale type, and sale category, be sure they can search for the booking directly as well.

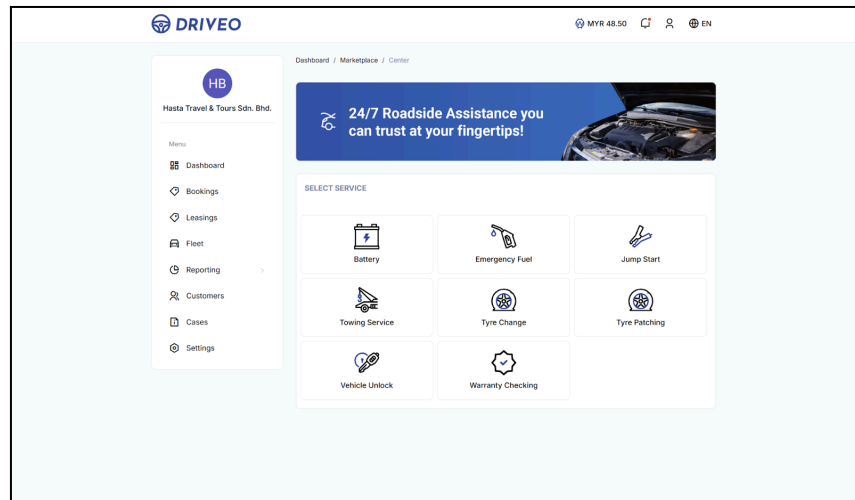


Figure 7: Maintenance Page of Current System

Figure 7 shows the available emergency maintenance and roadside assistance. Here, customers can call for roadside assistance due to problems with the battery, emergency fuel, jump start, tyre change and patching, and also towing service and vehicle unlocking.

Table 2 explains the current state of the car rental process, as it is a semi-digital and manual process, which involves human intervention and the use of semi-digital tools such as WhatsApp and bank transfer for communication between the company and the client. Most of the processes, including client identification, booking, and damage assessment, are done manually, which may cause inefficiency and errors. The current state of the system and the roles involved are as presented in Table 2.

Table 2: AS-IS System Functionality

Current Function	Role	Description
1. Registrati on	Customer	Customer required entering email and username, and creating a password.
2. Identity Verification	Staff	Staff verify documents manually.
3. Search Vehicles	Customer	Customer can search vehicles that he/she prefers and are available.
4. View Vehicle Details	Customer	Customer can know details of vehicles then decide which he/she wants.
5. Booking vehicles	Customer & Staff	Customer will fill in the booking form in Whatsform format and staff will fill the customer information into the system.
6. Payment method & Deposit	Customer	Customer complete the payment by transferring the amount of booking and deposit fees through the bank.
7. Booking confirmation	Staff	Staff confirm the customer's request after payment is successful.
8. Pickup & Handover	Staff & Customer	Staff confirm handover the vehicle to the customer.
9. Late	Staff	Staff check whether customer have

Current Function	Role	Description
Return Handling		successfully return the car before the return date. If not will have late return fees and notify the customer manually.
10. Return Checking & Damage Assessment	Staff	Check the car and records any damage, the damage cost will calculated manually
11. Deposit return	Staff	Deposit will be returned after the return of the car.
12. Cancellation & Refund Handling	Staff	Staff will handle this process based on the policy.
13. Blacklist & Risk Management	Admin, Staff	Users with unpaid fees, fraud, or repeated damages are flagged and restricted.
14. Admin Dashboard & Reporting	Admin	Able to monitor bookings, financial reports, car availability, demand trends and all customer information.
15. Feedback & Service Improvement	Customer, Staff, Admin	Customers submit ratings and comments after completing a rental and admins review to improve service quality.

8.2 Comparison Between Current and Existing System

1. DRIVEO (Current System)

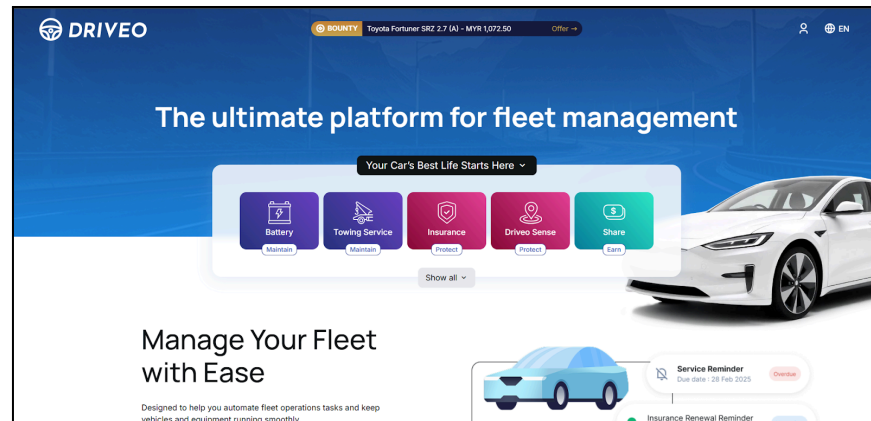


Figure 8: Homepage of DRIVEO

Figure 8 shows the DRIVEO system used by Hasta Travel to manage and handle their cars. The system lets staff or admin handle vehicle information, customer bookings information, payment records and car maintenance information. The system lacks integration with modern systems since it does not have real-time vehicle tracking, automated booking system and AI assistant for customer support. It results, the company having multiple major problems such as information overload and cluttered information.

2. WAHDAH (Existing System)

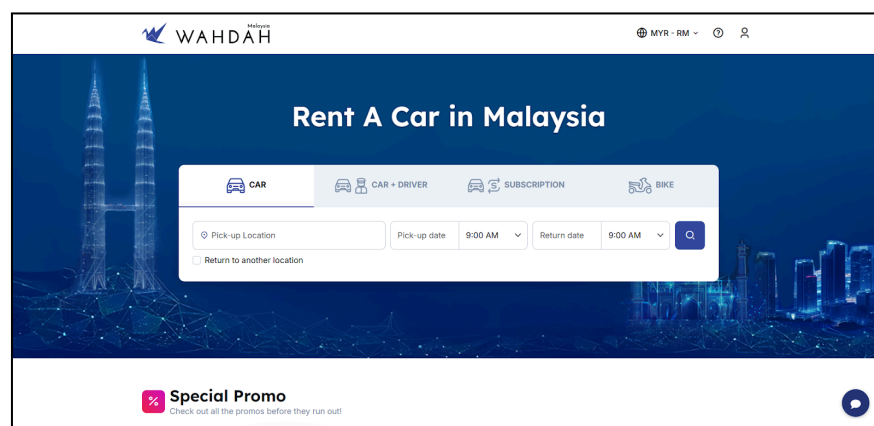


Figure 9: Homepage of WAHDAH

Figure 9 shows the WAHDAH website originated in Malaysia and provides a wide range of car rental services available for local and international travellers. Currently, the system is able to browse available cars, compare prices and complete payments directly on the website or mobile app. The system also has GPS for location tracking, insurance options to improve user experience and focus on user convenience.

3. SOCAR (Existing System)

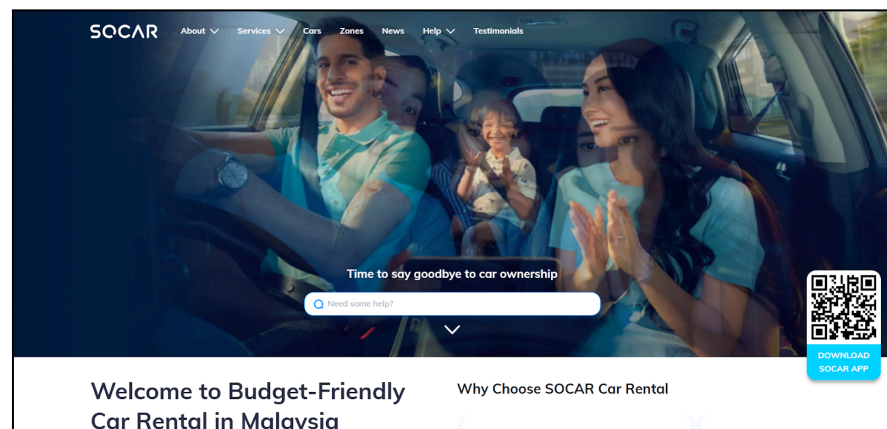


Figure 10: Homepage of SOCAR

Figure 10 shows SOCAR, a well-known car rental application in Malaysia because of its convenience system. SOCAR allows users to rent a car by the hour or day through a mobile application and offers vehicle availability in various locations. It has several features such as real-time car availability, keyless activation via mobile phone and an online payment system. SOCAR shows modern and convenient systems built to enhance the user experience of renting vehicles.

Table 3 shows a comparison of the current system and competitor system existing in the market right now. It compares multiple aspects such as platform accessibility, user roles & management, payment & deposit handling, vehicle tracking & status, booking management, customer management, notification system, security & authentication, analytics & reports, and customer support.

Table 3: Comparison Between Current System and Competitor Systems

System / Features	Current System	Competitor	
	DRIVEO	WAHDAH	SOCAR
Platform Accessibility	Website based platform	Website and mobile application	Mobile Application only
User Roles & Management	Limited control to staff (personalized for admin and customer)	Admin, Partner and customer able to access	User role only
Payment & Deposit Handling	Handled manually by staff	Online payment	In app payment and deposit system
Vehicle Tracking & Status	No GPS tracking & status updated manually	GPS tracking & car status system	Built in GPS and have trip logs
Booking Management	Online booking limitation, no instant availability	Real-time availability and instant confirmation	Real-time booking and smart keyless feature
Customer Management	Limited profile and no loyalty feature	Full profile and loyalty feature	Profile system
Notification System	Email updates only	Email and SMS updates	App push notifications

System / Features	Current System	Competitor	
	DRIVEO	WAHDAH	SOCAR
Security & Authentication	Limited data protection	Two-factor authentication and secure encryption	Account login with verification
Analytics & Reports	Basic booking records	Sales and performance analytics	Application usage statistics only
Customer Support	Manual communication	24/7 hotline support	In-app chatbot and helpdesk

9. Transaction Requirements

9.1 Data Entry

9.1.1 Core Transaction Data

- Customer Identity: Name, IC/Passport number, and details of the driving license.
- Timestamp: Date and time the vehicle is picked up and dropped off.
- Financials: The amount paid for the rental, the deposit paid, and the mode of payment (Cash, Bank Transfer, or QR).
- Reference Number: Invoice/receipt number for tracking the transaction.

9.1.2 Operational Requirements

- Vehicle ID: Vehicle plate number and model.
- Mileage Tracking: The odometer reading at the start and end of the trip to calculate the additional mileage charges.
- Fuel Level: To ensure the "Full-to-Full" or "Same-to-Same" policy is met.

9.1.3 Compliance & Audit Requirements

- Tax Requirements: To record SST (Sales & Service Tax) if applicable so you can report to LHDN.
- Audit Trails: If the car gets a speeding summons from PDRM/JPJ, your data entry will tell you who was using the vehicle at the exact time.
- Insurance Verification: In the event of an accident, the insurance company will need the transaction log to prove the vehicle was officially rented out to a licensed driver.

9.2 Data Update/Delete

The system is designed to allow staff and administrators to maintain a high level of data accuracy. Based on the system's operational logic, here are all the possible data updates that can be performed:

9.2.1 Vehicle & Fleet Maintenance

- **Availability Status:** Manually or automatically toggling a vehicle's status between "Available", "Booked", "Rented" and "In Maintenance".
- **Odometer Tracking:** Updating a vehicle's current mileage after every rental trip to ensure accurate usage logs.
- **Technical Specifications:** Modifying vehicle details such as rental rates (hourly/daily), color, or transmission type if the fleet is upgraded.
- **Document Renewal:** Updating vehicle insurance, road tax, and grant images when they are renewed annually.
- **Maintenance Logs:** Recording completed service dates, costs, and types.

9.2.2 User Account & Safety Management

- **Verification Workflow:** Changing a new user's status from "Pending" to "Verified" or "Rejected" after reviewing their documents.
- **Account Restrictions:** Updating a customer's status to "Blacklisted" or "Restricted" due to fraud, repeated damages, or unpaid fees.
- **Reason Documentation:** Recording and updating specific reasons for document rejection or account blacklisting to inform the customer.
- **Profile Maintenance:** Allowing users to update their personal contact information, home addresses, or security passwords.
- **Role Adjustments:** Administrators can update staff access levels or modify employee information within the management module.

9.2.3 Booking Lifecycle Management

- **Schedule Adjustments:** Modifying the pickup or return dates and times if a customer requests a rental extension.
- **Operational Stages:** Manually moving a booking through its lifecycle: "Pending Approval", "Confirmed", "Picked Up" and "Completed".

- **Cancellation Processing:** Marking a booking as “Cancelled” and updating the system to trigger refund or vehicle release logic.
- **Location Updates:** Changing the designated pickup or return location for specific booking records.
- **Remarks & Notes:** Adding internal staff remarks or special customer requests to an existing booking file.

9.2.4 Financial & Penalty Finalization

- **Payment Verification:** Updating the status of a rental payment from “Pending” to “Successful” once the receipt is verified.
- **Penalty Status:** Changing penalty or fine records from “Unpaid” to “Paid” after confirming the customer has settled the debt.
- **Fee Adjustments:** Adding or modifying extra charges such as fuel shortages, cleaning fees, or minor repair costs during the return inspection.
- **Deposit Release:** Updating the status of a security deposit to “Released” or “Refunded” after a successful car return.

9.2.5 Engagement & Loyalty Updates

- **Reward Point Balance:** Automatically updating a customer’s loyalty point total after a successful rental completion.
- **Voucher Status:** Updating a reward voucher’s status to “Used” or “Expired” once a customer redeems it.
- **Service Improvements:** Updating internal quality standards based on customer ratings and feedback comments.

9.2.6 Data Delete

- **Soft Delete Policy (Non-Physical Deletion)**
 - **Vehicles:** A vehicle is never permanently deleted. When a car is sold or retired, a Soft Delete is performed to remove it from the "active" view while preserving its records in the bookings and maintenances tables for historical Net Profit calculations.
 - **Payments & Agreements:** Payment and digital agreement records are never physically deleted. They are marked as

"Inactive" or "Archived" to maintain a complete audit trail for financial reports.

- **Physical (Hard) Deletion Rules**

- **Maintenance Logs:** Employees are permitted to Hard Delete a specific log entry in the maintenances table only if it was created by mistake or cancelled before any costs were incurred.
- **Unconfirmed Bookings:** Staff may physically delete a booking request if it has not yet been confirmed and no payment has been initiated.
- **Server Storage Management:** When updating vehicle or identity documents, the system deletes the old image files from the server storage only after the new files are successfully uploaded, preventing data bloat.

- **Integrity Constraints**

- **Referential Integrity:** The system prevents the deletion of any entity (e.g., Customer or Vehicle) that is currently linked to an active transaction in the bookings table.
- **Audit Logging:** Any update to core financial fields (amount, depoAmount) or user permissions is logged with the StaffID and a timestamp for security monitoring.

9.3 Data Queries

9.3.1 Customer Queries

- View available vehicles based on dates, times and vehicle types.
- View booking status and history.
- Review payment and deposit status.
- Check overdue payment and fines.
- Check loyalty points, stamps and redeemable vouchers.
- Monitor account verification status, including uploaded identification documents and license approvals.

9.3.2 Employee Queries

- Retrieve daily, weekly, monthly booking and customer records.
- Observe customer details and status verifications.
- Check vehicle availability, condition and status.
- Observe late return, possible fines and penalty records.
- Check loyalty and rewards status

9.3.3 Employer Queries

- Generate financial reports such as deposits, payments, fines and refunds.
- Observe general trends in bookings and how often vehicles are used.
- Obtain customer records and invoice statements.
- Manage staff information.

10. Benefits and Summary of Proposed System

10.1 System Overview

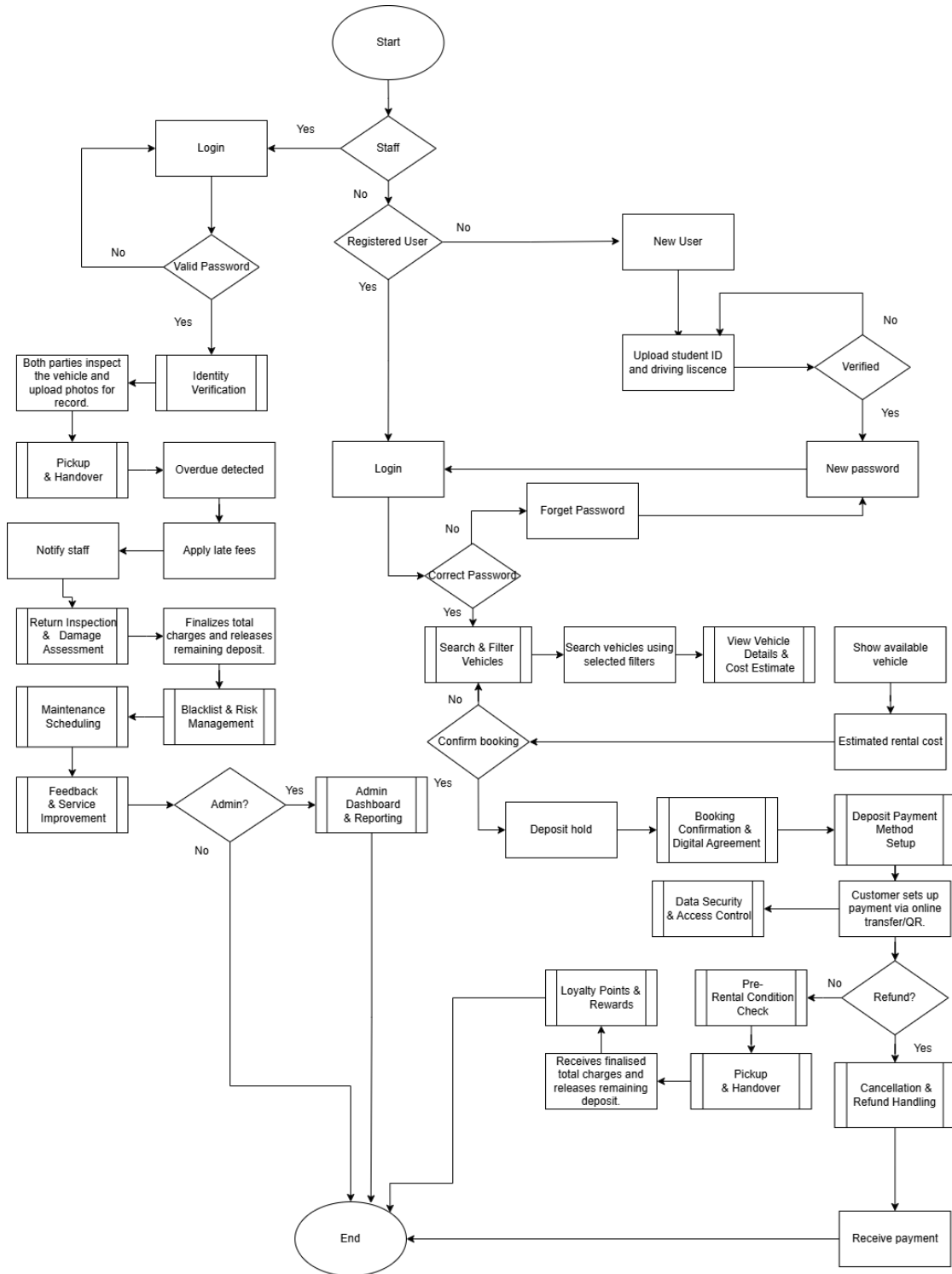


Figure 11: TO-BE System Workflow

Table 4: Overview of What the TO-BE System Will Do

Step	Role	Explanation
1. Registration	Customer	Signs up using a phone number or email and creates a password.
2. Role selection	Customer, Employee, Employer	Choose between the customer and the staff.
3. Profile & Document Upload	Customer	Completes personal profile and upload required documents which are student ID and driving license.
4. Identity Verification	Staff	Verifies documents manually or automatically. The customer account becomes “Verified” once approved.
5. Search & Filter Vehicles	Customer	Searches vehicles based on selected dates, location, vehicle type or price.
6. View Vehicle Details & Cost Estimate	Customer	The system displays details of available vehicles on filtered selection and the estimated rental cost.
7. Booking Request & Deposit Hold	Customer, System	Customer confirms booking, the system places a temporary deposit hold.
8. Booking Confirmation & Digital Agreement	System, Customer	System issues booking ID and digital rental contract for signature.
9. Deposit Payment Method Setup	Customer	The customer completes their payment method by online transfer or QR provided by the system.

Step	Role	Explanation
10. Pre-Rental Car Condition Check	Staff, Customer	Both inspect the vehicle, photos of current condition are uploaded for record.
11. Pickup & Handover	Staff, System	Staff confirms handover the vehicle to customer, system activates rental timer.
12. Late Return Handling	Staff, System	System auto-detects overdue return and applies late fees, notifying staff.
13. Return Inspection & Damage Assessment	Staff, System	Inspects the car and records any damage, system calculates repair cost if needed.
14. Payment Finalization & Deposit Release	System	Finalizes total charges and releases remaining deposit.
15. Cancellation & Refund Handling	System	Processes automatically based on policy.
16. Loyalty Points & Rewards	Customer, System	System awards points for completed rentals, users can redeem them for discounts on upcoming bookings.
17. Blacklist & Risk Management	Staff, Admin, System	Users with unpaid fees, fraud or repeated damages are flagged and restricted.
18. Admin Dashboard & Reporting	Admin	Able to monitor bookings, financial reports, car availability, demand trends and all customer information.
19. Maintenance Scheduling	Staff, System	The system schedules maintenance based on mileage or usage and updates vehicle availability.

Step	Role	Explanation
20. Data Security & Access Control	System	Payment receipt uploading , staff access is role-based to protect privacy
21. Feedback & Service Improvement	Customer, Staff, Admin	Customers submit ratings and comments after completing a rental and admins review to improve service quality.

According to **Figure 11** and **Table 4**, the proposed system would greatly simplify the entire vehicle rental process by streamlining all vehicle rental activities, including registration, reservation and payment into a single system. This is achieved by having customers sign up for an account, where they are required to upload verification documents for approval by staff. This would then allow customers to browse through available vehicles and reserve them. To enhance efficiency, the system would automatically generate electronic rental agreements, as well as carry out deposit pre-authorizations, in an effort to minimize paperwork from reservation to rental.

The transparency of the system is maintained through electronic recording of vehicle conditions during handover and return. The system would monitor the duration of rental in real-time, calculating late charges as well as final payments or deposit refunds after successful inspection. To encourage customer loyalty, the system would incorporate a points system, rewarding customers for repeated use.

From the point of view of the admin, the system utilizes a role-based access system for proper management and security. For the staff, they can carry out their daily activities, such as booking of supervisions and inspections of vehicles, while the administrators can keep track of the system through a comprehensive dashboard that shows them the financial reports, maintenance schedules and user risk management. This helps in the reduction of human error in the system, thus improving the reliability of the system.

10.2 USE Cases

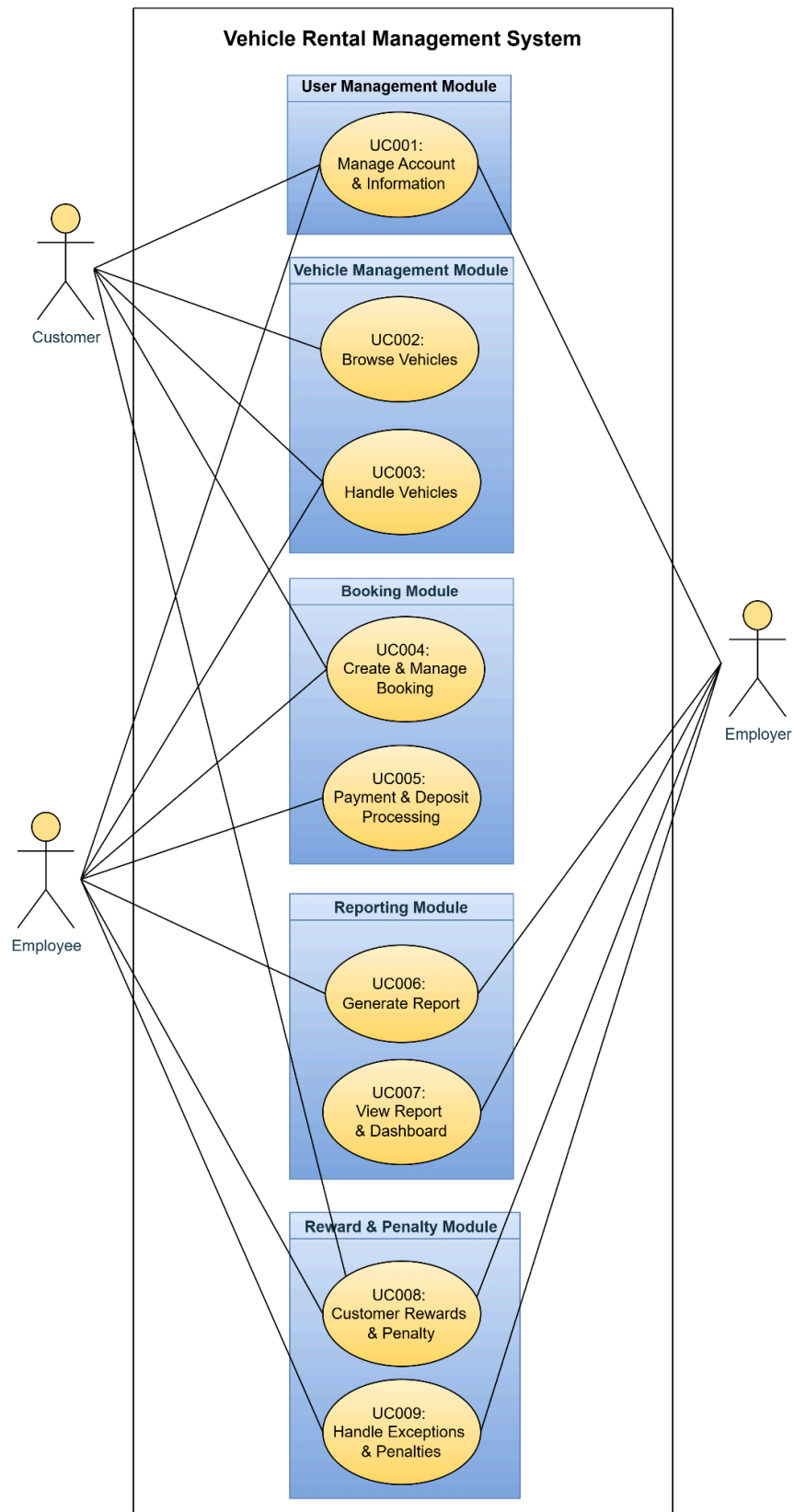


Figure 12: USE Case Diagram of TO-BE System

10.3 Workflow

1. User Registration & Verification

- 1.1. Customer creates an account, with an email or phone number, and then sets a password.
- 1.2 The customer fills their personal profile info and uploads the required documents such as a student ID and driving license.
- 1.3 Staff checks and verifies the documents uploaded.
- 1.4 Once approved, the customer will have a Verified status and then can make their bookings.

2. Searching & Booking

- 2.1 The customer searches for cars using filters including by date, location, vehicle type and price.
- 2.2 The system will display a list of available vehicles with determined rental costs.
- 2.3. The customer chooses the car and requests a booking.
- 2.4 The system will generate a booking ID and a digital rental agreement, which the customer will sign.
- 2.5 The customer configures their preferred payment through online transfer or QR payment.

3. Pre-Rental & Vehicle Handover

- 3.1 Staff and clients at the pickup check the car and attach photos for proof of the current condition of the car.
- 3.2 The staff confirms the handover of the vehicle and the rental timer will start counting as well as the customer using the car.
- 3.3 The system automatically detects late returns and applies the late fee, notifying the staff.

4. Return Process & Payment Closure

- 4.1 The staff inspects the car thoroughly when it is returned and notes any damages.
- 4.2 The system calculates total fees including repairs, late charges or fuel adjustments.

4.3 The system confirms the payment and the staff will releases any balance of the customer's deposit.

5. Cancellation & Refund (If Applicable)

5.1 The system automatically executes refunds in case of a cancellation stated in the agreement form.

6. Loyalty & Customer Engagement

6.1 During completion of the rental, customers are awarded loyalty points and claim the voucher after third time booking

6.2 Customers give feedback or ratings about their rental experience and the admin uses it to make improvements.

7. Staff & Admin Operations

7.1 The staff is responsible for verification, monitoring of booking, updating vehicle conditions and managing late returns.

7.2 The admins have access to a comprehensive dashboard that shows them bookings, financial reports, car availability, demand trends, and customer records.

7.3 The system automatically schedules vehicle maintenance based on mileage or usage and updates availability.

7.4 The system implements security through role-based access for the staff and admins.

7.5 Any violence, repeated damages or fees owed by a customer may be blacklisted by the staff or admin.

11. Summary

Hasta Vehicle Rental Management System is a solution of problem with an objective of solving the problem of changing the way in which Hasta Travel offers vehicle rental services in UTM. The analysis of the business situation has indicated a problem in which there are challenges in the process through WhatsApp and failure of the DRIVEO system in responding promptly, which has caused overlaps in the process. The main objective of this system is to create a web application that is not limited to any specific device and seeks to integrate the four major components of this system, which are the User module, Booking module, Fleet module, and Reporting module, in order to meet the potential customer base of 30,000 students and staff of the university.

The proposed system will be able to automate the entire rental process from the verification of the documents to the availability of the vehicle for rental. Some of the features that the proposed system will be able to offer include a staff dashboard for the maintenance of the vehicles and handing over of the vehicles, as well as a reward scheme for customer loyalty. With the inclusion of photo logging of the vehicles, the proposed system will be able to minimize the chances of errors that may be present in the current processing.

It is technically and operationally feasible, and this will ensure the creation of a safe environment with role-based access control and data management. This will allow the company to transition from the disjointed 'AS-IS' workflow process to the synchronized 'TO-BE' digital model. The 'TO-BE' model will allow the company to make data-driven decisions with the help of advanced admin analytics. This will allow Hasta Travel to conform to the standards set by the modern market, which are represented by the likes of SOCAR and WAHDAH.

12. Glossary

1. AI (Artificial Intelligence)

The ability of computer systems to perform tasks that normally require human intelligence, such as learning, problem-solving, and decision-making.

2. AS-IS Process

The current state of a business process or system before any improvements or changes are made.

3. Device-Agnostic

Hardware-independent software that is compatible with various operating systems and devices, such as mobile phones and computers.

4. DRIVEO

The current online fleet management system used by Hasta Travel for record-keeping and basic booking.

5. Fleet

The collection of vehicles (economical hatchbacks, sedans, SUVs, and MPVs) owned and operated by the rental company.

6. GPS (Global Positioning System)

A satellite-based navigation system that provides location, navigation, and timing information.

7. ID (Identity Document)

A document proving a person's identity, specifically a Student ID or Driving License in this system.

8. JPJ (Jabatan Pengangkutan Jalan)

The Road Transport Department of Malaysia, responsible for vehicle registration and licensing.

9. LHDN (Lembaga Hasil Dalam Negeri)

The Inland Revenue Board of Malaysia, responsible for tax collection and reporting.

10. PDRM (Polis Diraja Malaysia)

The Royal Malaysia Police, who handle traffic summons and accident reports.

11. QR Code (Quick Response Code)

A machine-readable code used in this system for processing digital payments and transfers.

12. RDBMS (Relational Database Management System)

A type of database management system that stores data in tables and maintains relationships between them.

13. SMS (Short Message Service)

A text messaging service that allows the exchange of short text messages between mobile devices.

14. SST (Sales & Service Tax)

A consumption tax in Malaysia applied to specific services, including car rentals.

15. TO-BE Process

The proposed future state of a business process or system after digital transformation and automation.

16. UAT (User Acceptance Testing)

The final phase of the software testing process where the system is tested in the “real world” by the intended audience.

17. UTM (Universiti Teknologi Malaysia)

The host institution and primary community served by Hasta Travel & Tours Sdn. Bhd.

18. WBL (Work-Based Learning)

An educational strategy that provides students with real-life work experiences where they can apply academic and technical skills.

19. WBS (Work Breakdown Structure)

A deliverable-oriented breakdown of a project into smaller, manageable components.

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